Week 6, July-Sept 2017
MOOCS Decision Modeling Course

Simulation

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Prof. Biswajit Mahanty

Professor

Department of Industrial and Systems Engineering

Indian Institute of Technology Kharagpur

Simulation Examples (Continued)

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Single Server Example

In a single server bank counter, inter-arrival time of the customers as well as the service time of the bank personnel are in accordance with a probability distribution not estimated. 200 occurrences each of inter-arrival times and service times are noted to be as follows:

Interarrival Time	Occurrence		Service Time	Occurrence
3 minutes	30		4 minutes	8
6 minutes	50		6 minutes	20
9 minutes	74		8 minutes	36
12 minutes	32		10 minutes	88
15 minutes	14		12 minutes	48

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Using random numbers, simulate the queuing system for 10 incoming customers. Find the average waiting time per customer in the queue. Assuming the simulation starts at 9.00 AM, find out the time intervals in which there are nobody in the queue.

Random Numbers for interarrival time: 58, 47, 23, 69, 35, 55, 69, 90, 86, 74

Random Numbers for service time: 87, 39, 28, 97, 69, 87, 52, 52, 15, 85

Random Number Allocation

Interarrival Time

Interarrival Time	Occurrence (Total = 200)	Probability (Occurrence/200)	Cumulative Probability	Random Nos Allocated
3 minutes	30	0.15	0.15	00-14
6 minutes	50	0.25	0.40	15-54
9 minutes	74	0.37	0.77	55-76
12 minutes	32	0.16	0.93	77-92
15 minutes	14	0.07	1.00	93-99

Service Time

Service Time	Occurrence (Total = 200)	Probability (Occurrence/200)	Cumulative Probability	Random Nos Allocated
4 minutes	8	0.04	0.04	00-03
6 minutes	20	0.10	0.14	04-13
8 minutes	36	0.18	0.32	14-31
10 minutes	88	0.44	0.76	32-75
12 minutes	48	0.24	1.00	76-99

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Simulation for 10 Arrivals

SL.	Random No.	Inter-Arrival Time	Arrived At	Random No.	Service Time	Service Begins At	Service Ends At	Waiting Minutes
1	58	9	9.09	87	12	9.09	9.21	00
2	47	9	9.18	39	10	9.21	9.31	03
3	23	6	9.24	28	8	9.31	9.39	07
4	69	9	9.33	97	12	9.39	9.51	06
5	35	6	9.39	69	10	9.51	10.01	12
6	55	9	9.48	87	12	10.01	10.13	13
7	69	9	9.57	52	10	10.13	10.23	16
8	90	12	9.09	52	10	10.23	10.33	14
9	86	12	10.21	15	8	10.33	10.41	12
10	74	9	10.30	85	12	10.41	10.53	11

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- **Total Waiting Time in queue for 10 customers = 94 minutes**
- **Hence, Average Waiting Time per customer in queue = 94/10 = 9.4 minutes.**
- **So, Time intervals when there were nobody in the queue: 9.21 to 9.24, and 9.31 to 9.33.**

M/M/1 Simulation Example

An example from Queuing: In a bank with a single counter, Arrival is Poisson at 5 per hour and the Service time is exponential at 10 minutes per service.

Hence, Arrival Rate λ = 5 per hour

Service Rate μ = 60/10 = 6 per hour

How to simulate this queuing system?

- **Generate inter-arrival and service times from exponential distribution – how? Use random variate**
- **Make use of Fixed or Variable Time Increment Method to obtain and record events.**
- **From the simulation data generated, obtain steady state parameters such a L, Lq, W, and Wq.**

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Discrete Event Simulation Example

Queuing System

- Entities: Customers, Servers, Bank
- Attributes: Arrival rate, Service rate, Queue space etc.
- Activities: Arrival, Service
- Events: An arrival, A Service Completed.
- State Variables: Number of Customers in the System, Number of Customers in Queue

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M/M/1 Simulation Example

R1	tarr = – ln(R1)/arate	R2	tser = – ln(R2)/srate
0.918	1	0.795	2
0.454	9	0.715	3
0.921	1	0.024	37
0.349	13	0.537	6
0.916	1	0.485	7
0.827	2	0.251	14
0.238	17	0.687	4
0.582	6	0.400	9
0.435	10	0.744	3
0.275	15	0.622	5
0.526	8	0.851	2
0.343	13	0.112	22
0.580	7	0.142	20
0.906	1	0.173	18
0.139	24	0.407	9
0.334	13	0.261	13
	arate 5/hr		srate 6/hr

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Generating Arrival and Service

R1	tarr = – ln(R1)/arate	R2	tser = – ln(R2)/srate	Arrived at	Service at
0.918	1	0.795	2	1	3
0.454	9	0.715	3	10	13
0.921	1	0.024	37	11	50
0.349	13	0.537	6	24	56
0.916	1	0.485	7	25	63
0.827	2	0.251	14	27	77
0.238	17	0.687	4	44	81
0.582	6	0.400	9	50	90
0.435	10	0.744	3	60	93
0.275	15	0.622	5	75	98
0.526	8	0.851	2	83	100
0.343	13	0.112	22	96	122
0.580	7	0.142	20	103	142
0.906	1	0.173	18	104	160
0.139	24	0.407	9	128	169
0.334	13	0.261	13	141	182
	arate 5/hr		srate 6/hr		

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Waiting Time Computation

R1	tarr = – ln(R1)/arate	R2	tser = – ln(R2)/srate	Arrived at	Service at	Waiting Time	Waiting Time in Queue
0.918	1	0.795	2	1	3	2	0
0.454	9	0.715	3	10	13	3	0
0.921	1	0.024	37	11	50	39	2
0.349	13	0.537	6	24	56	32	26
0.916	1	0.485	7	25	63	38	31
0.827	2	0.251	14	27	77	50	36
0.238	17	0.687	4	44	81	37	33
0.582	6	0.400	9	50	90	40	31
0.435	10	0.744	3	60	93	33	30
0.275	15	0.622	5	75	98	23	18
0.526	8	0.851	2	83	100	17	15
0.343	13	0.112	22	96	122	26	4
0.580	7	0.142	20	103	142	39	19
0.906	1	0.173	18	104	160	56	38
0.139	24	0.407	9	128	169	41	32
0.334	13	0.261	13	141	182	41	28
	arate 5/hr		srate 6/hr		Sum	517 min	343 min
					Avg	57.44	38.11

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Comparing with Theoretical Values

Now, let us compare the results with theoretical values:

- **Arrival rate: $\lambda = 5$/hour** **Service rate: $\mu = 6$/hour**
- **Hence utilization $\rho = \lambda/\mu = 5/6$**
- **So, Average Waiting Time $W = L/\lambda = (\rho/(1-\rho))/\lambda = ((5/6)/(1/6))/5 = 1$ hour = 60 minutes**
- **Average Waiting Time in queue $W_q = L_q/\lambda =$**

$(\rho^2/(1-\rho))/\lambda = ((25/36)/(1/6))/5 = 5/6$ hour = 50 minutes

	Waiting Time (W)	Waiting Time in Queue (Wq)
Theoretical Value	60 Minutes	50 minutes
Simulated Value	57.44 minutes	38.11 minutes

What about sensitivity of the simulation results?

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Another Replication

R1	tarr = – ln(R1)/arate	R2	tser = – ln(R2)/srate	Arrived at	Service at	Waiting Time	Waiting Time in Queue
0.487	9	0.207	16	9	25	16	0
0.177	21	0.965	0	30	30	0	0
0.242	17	0.432	8	47	55	8	0
0.125	25	0.397	9	72	81	9	0
0.325	13	0.504	7	85	92	7	0
0.162	22	0.375	10	107	117	10	0
0.249	17	0.226	15	124	139	15	0
0.037	40	0.001	65	164	229	65	0
0.727	4	0.668	4	168	233	65	61
0.252	17	0.076	26	185	259	74	48
0.775	3	0.231	15	188	274	86	71
0.523	8	0.637	5	196	279	83	78
0.323	14	0.661	4	210	283	73	69
0.760	3	0.164	18	213	301	88	70
0.620	6	0.663	4	219	305	86	82
0.348	13	0.506	7	232	312	80	73
	arate 5/hr		srate 6/hr		Sum	765	552
					Avg	85.00	61.33

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Monte-Carlo Simulation and Output Analysis

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Monte Carlo Simulation

Find an approximate value of $\sqrt{5}$ with the help of 5000 random numbers from a uniform distribution over (0,1).

Answer:

- **Obtain 5000 random numbers r_i's uniformly distributed between 0 and 1**
- **Transform the random numbers to uniformly distributed random numbers x_i's between 2 and 3 by:**

  $x_i = r_i + 2$ **Why? because** $2^2 <= (\sqrt{5})^2 <= 3^2$

- **Obtain x_i^2 and count the number p for which we have:**

  $x_i^2 < 5;$

- **Now, Approximate value of $\sqrt{5}$ will be 2 + p/5000 (~2.236)**
- **Why does it happen?**

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Monte Carlo Simulation

- ***A problem on Prisoners' Dilemma***

- ***Two prisoners A and B are charged with a joint crime and are held incommunicado. If A confesses and B does not, A is given a reduced sentence of two years for cooperating with the authorities and B gets a 10 year prison term, and vice versa. If both confesses, each gets six years. If neither confesses, each will receive only a one-year sentence. Neither can know or even control the behavior of the other.***

- ***Design a simulation scheme in order to obtain the expected penalties of A and B under varying probabilities of prisoners taking different actions***

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Monte Carlo Simulation

Let us assume the following probabilities:

- **A confesses : a**
- **A does not confess : 1 – a**
- **B confesses : b**
- **B does not confess : 1 – b**

Choose 2 random numbers r_1 and r_2 between 0 and 1.

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If r_1 <= a;
Then *A confesses*
Else
A does not confess

If r_2 <= b;
Then *B confesses*
Else
B does not confess

Monte Carlo Simulation

- Depending on the outcomes, one of the following events will come about and corresponding penalties would result:

Event	Penalty
A confesses but B does not confess	A: 2 years and B: 10 years
A does not confess but B confesses	A: 10 years and B: 2 years
A confesses and B also confess	A: 6 years and B: 6 years
A does not confess and B does not confess	A: 1 year and B: 1 year

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- Finally, the expected penalty values for A and B are the average values of the corresponding penalties obtained in different iterations.

Monte Carlo Simulation

- As a check, these expected penalty values can be compared with the following theoretically obtained values:

Prisoner	Expected Penalty
A	a*(1 – b)*2 + (1 – a)*b*10 + a*b*6 + (1 – a)*(1 – b)*1 years
B	a*(1 – b)*10 + (1 – a)*b*2 + a*b*6 + (1 – a)*(1 – b)*1 years

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Length of Simulation Run

- Estimation of Length of Simulation run is a non-trivial job.
- It depends on confidence level desired, acceptable tolerance limit, and variance of parent population.
- It is to note that average value of the random variable being estimated is normally distributed with a variance equal to σ^2/n

$$\sigma / \sqrt{n} = (t / y_{1-\alpha/2}) \quad \text{or} \quad y_{1-\alpha/2} = (t/\sigma) * \sqrt{n}$$

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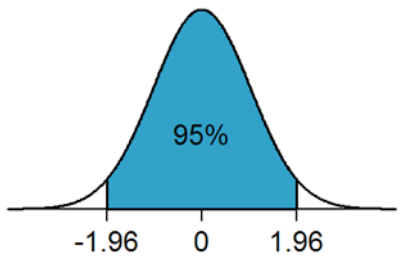

Where, t : tolerance limit we are ready to accept

σ^2 : variance of the parent population

$(1 - \alpha)$: specified value of confidence level

$y_{1-\alpha/2}$: 2-tailed standardized normal statistic for prob $(1-\alpha)$

n : Length of simulation run or number of samples

Length of Simulation Run

Impact of variance reduction on Length of simulation run

Length of Simulation run is given by: $n = \sigma^2 * (y_{1-\alpha/2}/t)^2$

Where, t : tolerance limit we are ready to accept

σ^2 : variance of the parent population

$y_{1-\alpha/2}$: 2-tailed standardized normal statistic for prob $(1-\alpha)$

- **It can be seen that the variance of the parent population, σ^2 is directly proportional to the length of the simulation run n.**
- **Thus, if variance σ^2 reduces, the required length of simulation run will also reduce.**
- **Reduction in the length of simulation run is desired to make the simulation more efficient as it is less time consuming.**

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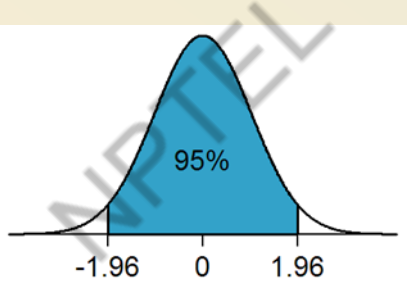
Length of Simulation Run Example

In a company, weekly production has a random variation with a standard deviation of 100. Find length of simulation run for estimating average value of weekly production within ±20 units with a confidence level of 95%.

Note: Standardized normal statistic for 95% probability is 1.96



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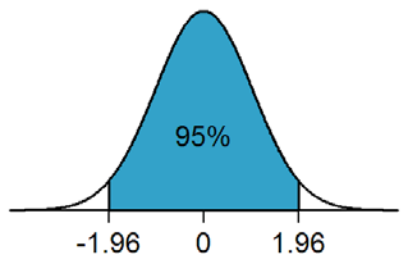
Length of Simulation Run Example

In a company, weekly production has a random variation with standard deviation of 100. Find length of simulation run for estimating average value of weekly production within ±20 units with a confidence level of 95%. *Note: Standardized normal statistic for 95% probability is 1.96*



Length of Simulation run n may be obtained from:

where:

$$y_{1-\alpha/2} = (t/\sigma) * \sqrt{n}$$

(1 – α): specified value of confidence level = 0.95

$y_{1-\alpha/2}$: 2-tailed standardized normal statistic for 0.95 = 1.96

- **t : tolerance limit we are ready to accept = 20**
- **σ : s.d. of the parent population = 100**
- **n : Length of simulation run or number of samples required.**

Hence 1.96 = (20/100)*√n; So, n = $(1.96*100/20)^2$ = 96

So, the length of the simulation run should be at least 96 weeks for which weekly production need to be generated.

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Variance Reduction and Simulation Software

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Variance Reduction Techniques

- Antithetic sampling helps in variance reduction in discrete event simulation
- In Antithetic sampling, negative correlation is introduced between pairs of observations in a simulation experiment for obtaining more statistically reliable answer.
- How Antithetic sampling works?
- While estimating mean value of a random variable, instead of simulation run of length n, two simulation runs each of length n/2 are considered with values of the observations in the two runs as y_i's and z_i's.
- Unbiased estimate of grand mean $\bar{x}$ would be $(\bar{y}+\bar{z})/2$
- The variance of this grand mean $\bar{x}$ of the two replications:

$$\sigma_{\bar{x}}^{2} = \frac{(\sigma_{\bar{y}}^{2} + \sigma_{\bar{z}}^{2}) + 2 * Cov(\bar{y}, \bar{z})}{4}$$

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Variance Reduction Techniques

- **Antithetic sampling - estimating mean value of a random variable - two simulation runs of length n/2 each are considered instead of one simulation run of length n, wth values of the observations in the two runs as y_i's and z_i's.**
- **Unbiased estimate of grand mean $\bar{x}$ is $(\bar{y}+\bar{z})/2$**
- **The variance of this grand mean $\bar{x}$ of the two replications:**

$$\sigma_{\bar{x}}^{2} = \frac{(\sigma_{\bar{y}}^{2} + \sigma_{\bar{z}}^{2}) + 2 * Cov(\bar{y}, \bar{z})}{4}$$

- **With negative correlation between each of the n/2 pairs of observations, covariance term will be negative**
- **So, variance of the mean of n/2 pairs of observations will be less than variance of the mean of one continuous run of n observations!**

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Antithetic Sampling Example

- Negatively co-related pairs of samples can be generated by performing two identical simulation runs but with different sets of uniformly distributed random numbers.
- First set may use any set of random numbers, $r_1, r_2, r_3, \ldots r_{n/2}$.
- The second set, then, should use random numbers,

  $(1-r_1), (1-r_2), (1-r_3), \ldots (1-r_{n/2})$.
- Thus if r_1 is greater than the average value, $(1-r_1)$ will be less than the average value and vice versa.
- Two sets of random numbers will be negatively correlated.
- Hence, corresponding outputs will also tend to be negatively correlated and the variance of the grand mean would be suitably reduced.

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Antithetic Sampling Example

- In a single server queue, inter-arrival time and time spent in the system are negatively correlated - if inter-arrival time is short, customers are in the queue and spend longer time.
- Also, service time and time spent by customers in system are positively correlated. Quicker the service, less is time spent.
- Thus, inter-arrival time and service time have opposite effects on the time a customer spends in the system. Thus inter-arrival time and Service time are correlated negatively.
- Thus, for variance reduction, the following may be adopted:

One set of Random Numbers to generate	Another Set of Random Numbers to generate
• inter-arrival times in 1st replication • service times in 2nd replication	• service times in 1st replication • inter-arrival time in 2nd replication

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Verification and Validation

Verification

- Is the model built right?
- Does it serve the intended purpose?
- Computational model should be consistent with specification model

Validation

- Is the right model being built?
- Computational model should be consistent with the real system under consideration – does the model adequately describe the reality being modeled?
- Simulation output and system output should match – the decision makers must trust the model built.

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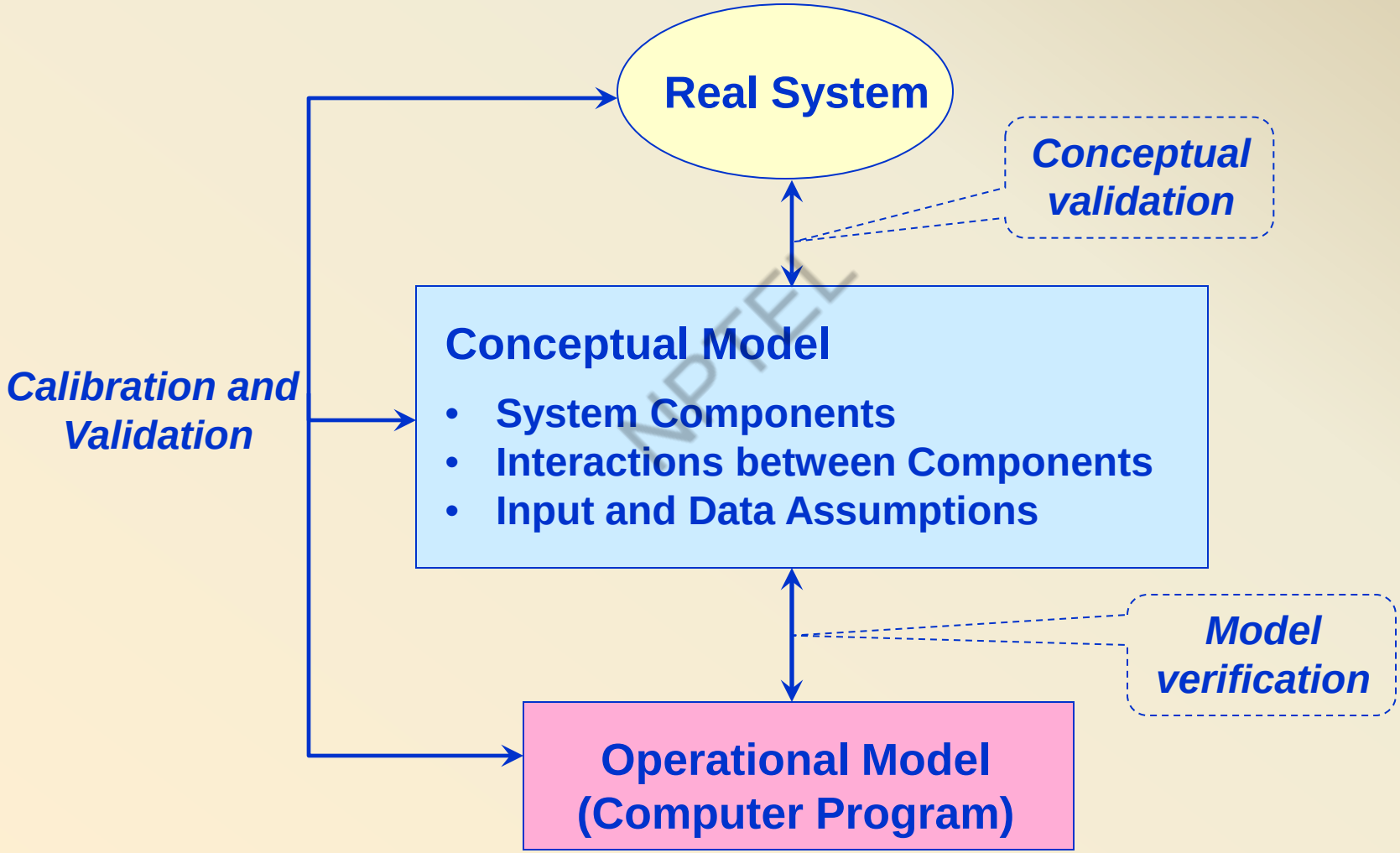
Verification and Validation



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Arena Simulation Software

- Arena discrete-event simulation software uses flowchart modeling methodology with a large library of pre-defined building blocks to model system processes.
- It uses stochastic systems to generate random numbers with a range of statistical distribution options to model process variability.
- Multiple simulation runs or replications can be generated using the software.
- Arena translates the user's actions into the SIMAN simulation language code to execute the model.
- The output of the simulation is an estimate of the true system behavior.

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Arena Simulation Blocks

- **CREATE: models arrival where entities sequentially enter the model in a specified pattern.**
- **QUEUE: provides waiting space for the entities whose movements are suspended based on system status.**
- **SEIZE: models entities from previous queue block to enter the seize block and seize idle resource.**
- **DELAY: models consumption of time of entities as they are allocated resources.**
- **RELEASE: models entities releasing resources when activity requiring resources are completed.**
- **COUNT: counts number of occurrences of some event.**

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Arena Simulation Blocks

- **DISPOSE: provides a mechanism for modeling the departure of entities from the system.**
- **GROUP: groups a set of entities having a particular set characteristics.**
- **SPLIT: splits or de-groups the entities that had been grouped earlier before they are disposed.**
- **ASSIGN: assigns values to attributes and variables used in a model.**
- **BRANCH: directs entities to different sections of the model depending upon true or false conditions.**

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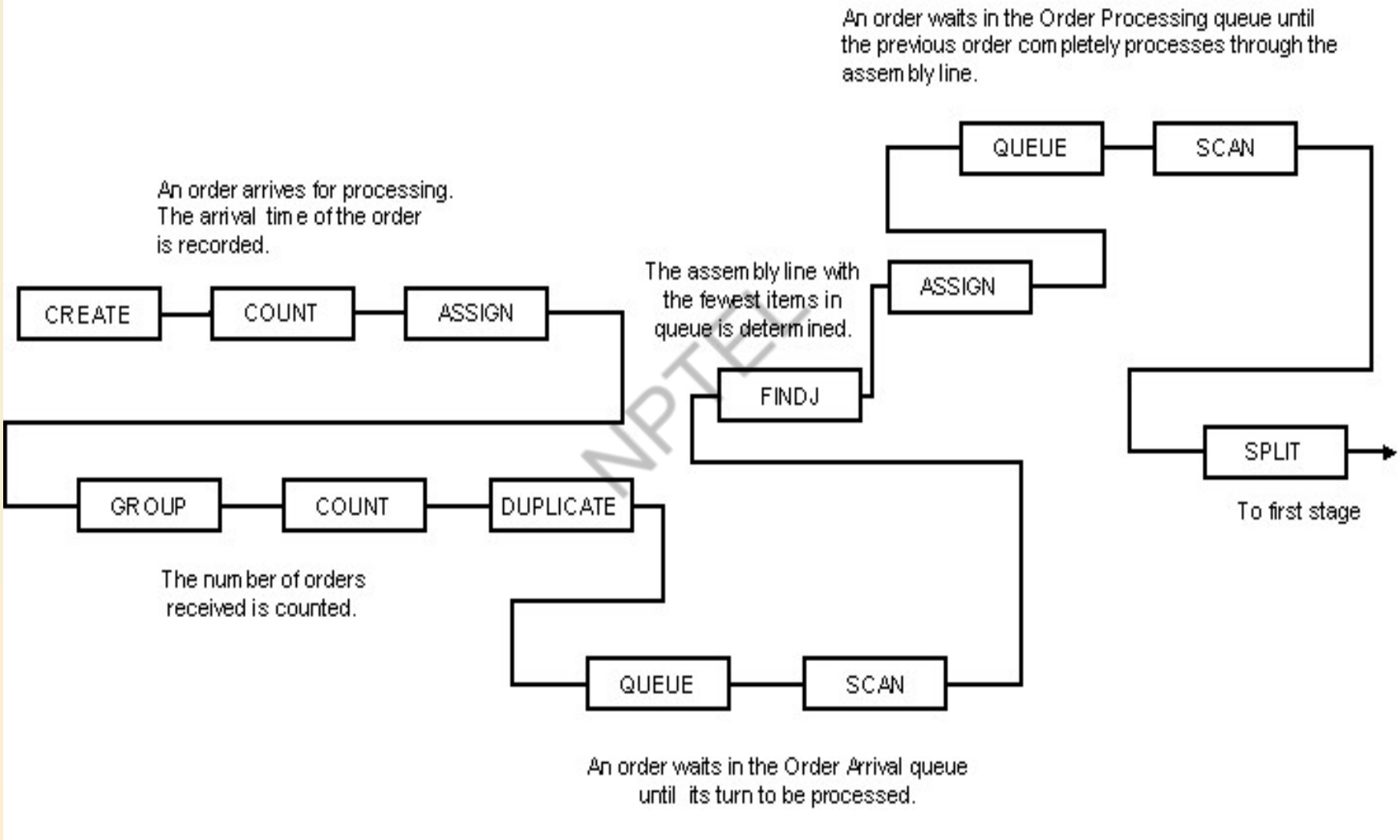
Arena Simulation Example



Reference: http://www.uh.edu/~lcr3600/simulation/models.html

Module 29

Continuous Simulation and System Dynamics

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Continuous Simulation Models

Continuous Simulation Models

- ***Deterministic or Stochastic***

  Deterministic or random state variables

- ***Dynamic***

  State changes with time

- ***Continuous***

  State changes continuously
  guided by a set of differential equations

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Continuous Simulation Models

Continuous Simulation Models

- Events do not occur
- Changes expressed in terms of differential equations
- System state evolves continuously in time
- Random Numbers are not required
- Examples:, Classical Mechanics, Econometric models, System Dynamics

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Classical Mechanics Example

Velocity v

Displacement s

Acceleration a

We know that

Acceleration dv(t)/dt = a(t) and Velocity ds(t)/dt = v(t)
Displacement = s(t)

Using Difference Equations, We can write:

v(t+dt) = v(t) + dt*a(t) + higher order terms
s(t+dt) = s(t) + dt*v(t) + higher order terms

Input: Acceleration Profile a(t), a(t+dt), a(t+2dt) etc.

Can we generate velocity and displacement profiles from these equations by simulation?

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Classical Mechanics Example

Velocity v

Displacement s

Acceleration a

Acceleration dv(t)/dt = a(t) and Velocity ds(t)/dt = v(t)

Using Difference Equations, We can write:

v(t+dt) = v(t) + dt*a(t) + higher order terms
s(t+dt) = s(t) + dt*v(t) + higher order terms

Input: Acceleration Profile a(t), a(t+dt), a(t+2dt) etc.

Starting with a(t), v(t), and s(t) values, say for t=0,
For small dt and neglecting higher order terms,

We can find out values for

vt+dt), v(t+2dt)… as well as s(t+dt), s(t+2dt)…. etc. etc.

Thus, we can generate velocity and displacement profiles through simulation.

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Distributed Lag Models

- These models are continuous simulation models. The properties change only at fixed intervals of time
- Current values of the variables are based on current values of other relevant variables or on values of the variables that have occurred in previous intervals.
- Used in econometric studies where uniform steps are related to time intervals over which data is collected.
- Consist of linear, algebraic equations.

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Distributed Lag Models

A model of national economy

- C: Consumption I: Investment T: Taxes
- G: Government Expenditure N: National Income

Equations:

- $C = 30 + 0.8*(N_{-1} - T_{-1})$ N_{-1} refers to one period old value of N
- $I = 3 + 0.2*N_{-1}$
- $N = C_{-1} + I_{-1} + G_{-1}$
- $T = 0.3*N$ No lag present in this equation

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- Simulation needs initial values C, I, T, G, and N at time = 0
- Variables at Time 1 can be obtained using the equations. Then Time 2 values etc. Variables can be plotted over time.
- Further needs are sensitivity of initial values, implementing policy changes, and generation of alternate scenarios.

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Distributed Lag Models

A model of national economy

- C: Consumption I: Investment T: Taxes
- G: Government Expenditure N: National Income
- $C = 30 + 0.8*(N_{-1} – T_{-1})$ N_{-1} refers to one period old value of N
- $I = 3 + 0.2*N_{-1}$
- $N = C_{-1} + I_{-1} + G_{-1}$
- $T = 0.3*N$ No lag present in this equation
- C_{-1} I_{-1} T_{-1} G_{-1} N_{-1} - one period old values are known
- Since Government Expenditure Profile is known, using one period old values, find C, I, T, N values.
- Advance time, Rename C, I, T, N as C_{-1} I_{-1} T_{-1} G_{-1} N_{-1}
- Using them, calculate next period values and so on.

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System Dynamics Models

System Dynamics is a methodology of social systems analysis and design at the aggregate level using continuous simulation.

System Dynamics was developed in late 1950s by Jay W Forrester at Sloan School of Management, MIT.

System Dynamics

- addresses long-term issues of real systems
- models complex non-linear relationships with ease
- models feedback processes
- models soft social and psychological variables
- tests effects of alternative policy options
- communicates results and recommendations with ease

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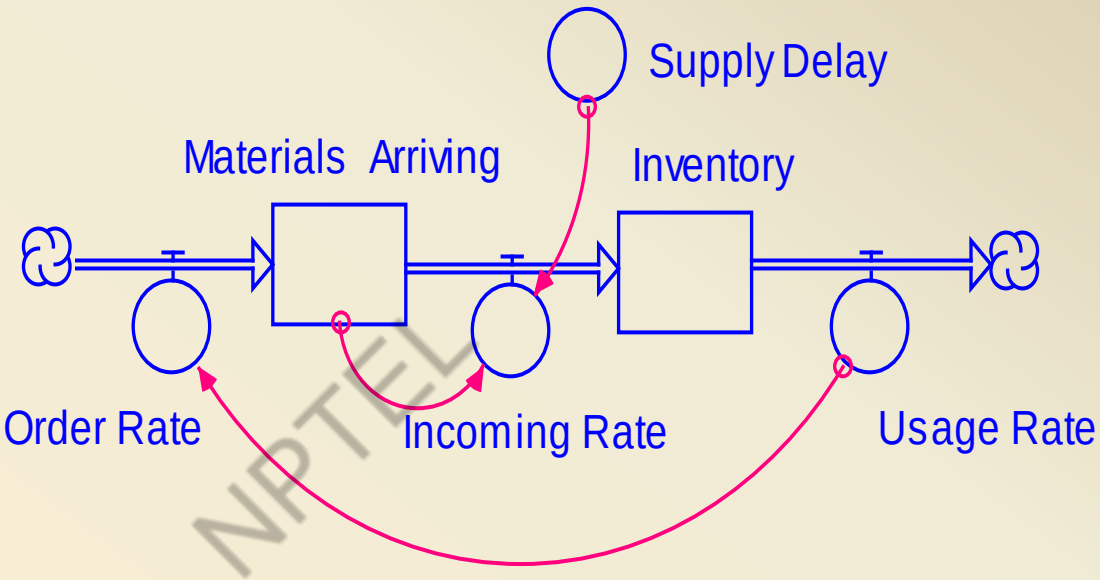
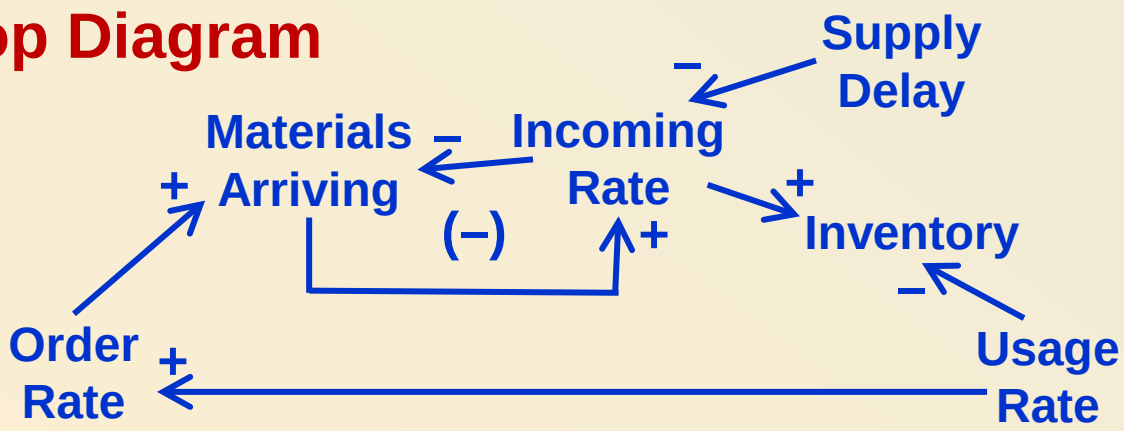
System Dynamics Models

Flow Diagram



Causal Loop Diagram



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Cause and Effect

Assuming a steady state condition for the model, when a change is brought about in A, and if a resulting change comes in an immediate and adjacent variable B, then:

<u>A (causal variable) causes B (affected variable)</u>

Causality due to Conservation consideration

Production → + Inventory

Depreciation → − Capacity

Birthrate → + Population

Direct Causality

Quality → + Sales

Price → − Demand

Skill → + Productivity

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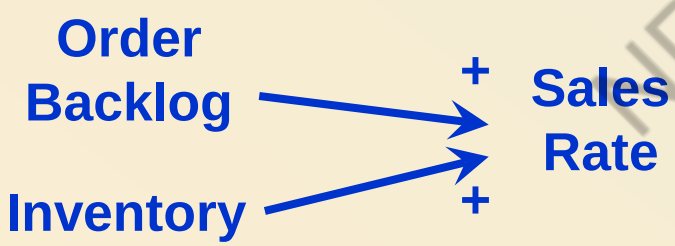
Cause and Effect

Examples of Causal Chain

A) Price → ⁻ Demand → ⁺ Desired Production → ⁺ Inventory

B) Skill → ⁺ Productivity → ⁺ Production → ⁺ Inventory

Multivariate Causality



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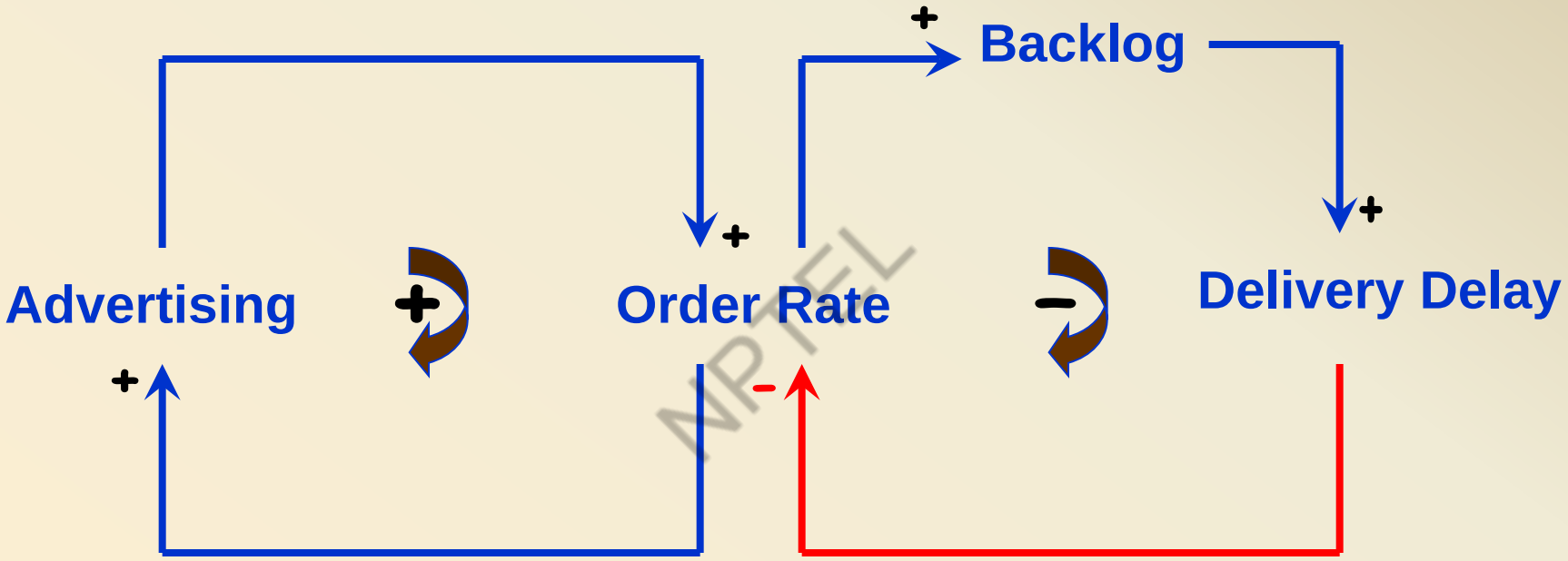
Dynamics of Customer Orders



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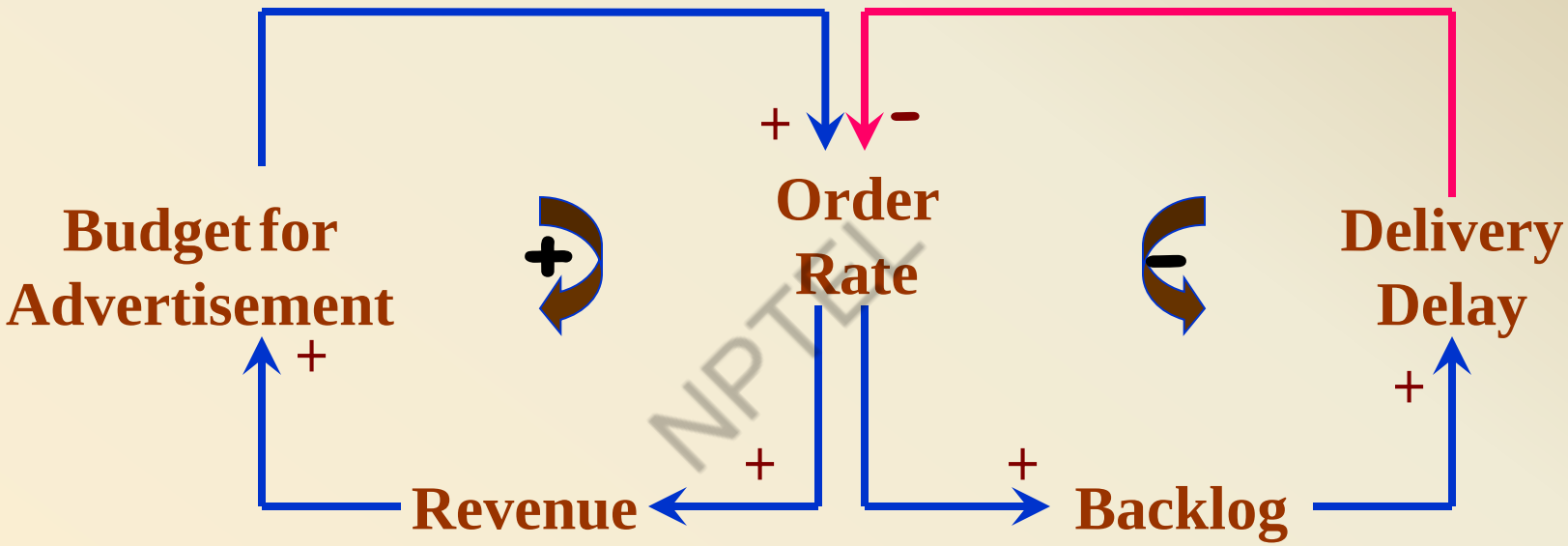
Interacting Causal Loops



Causal Loops and Multivariate Causality give rise to Interacting Causal Loops

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System Dynamics Model Examples

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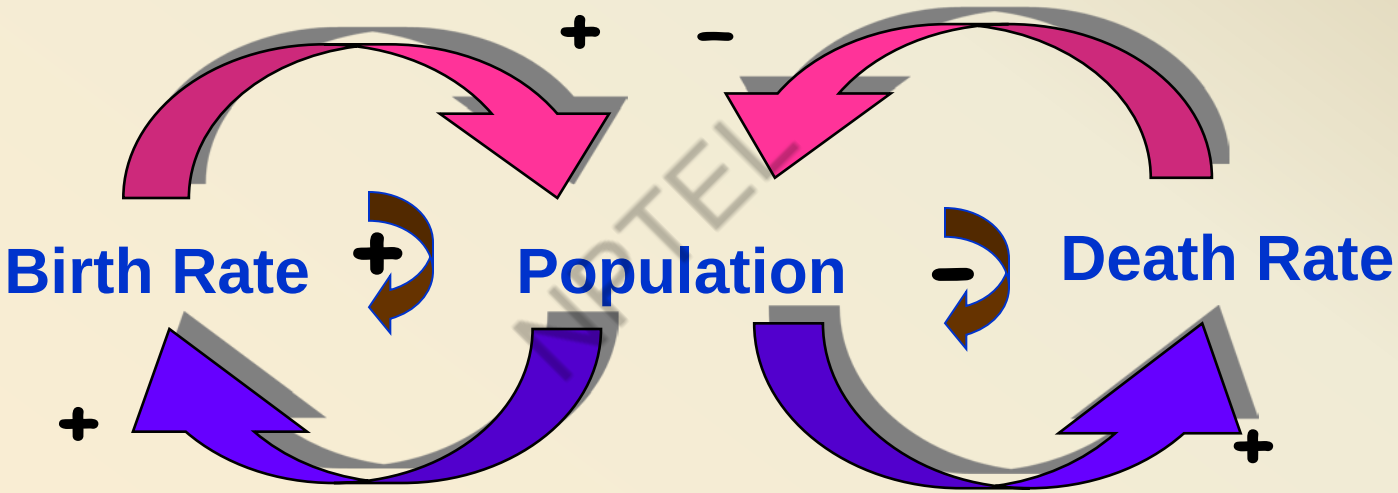
Population Dynamics



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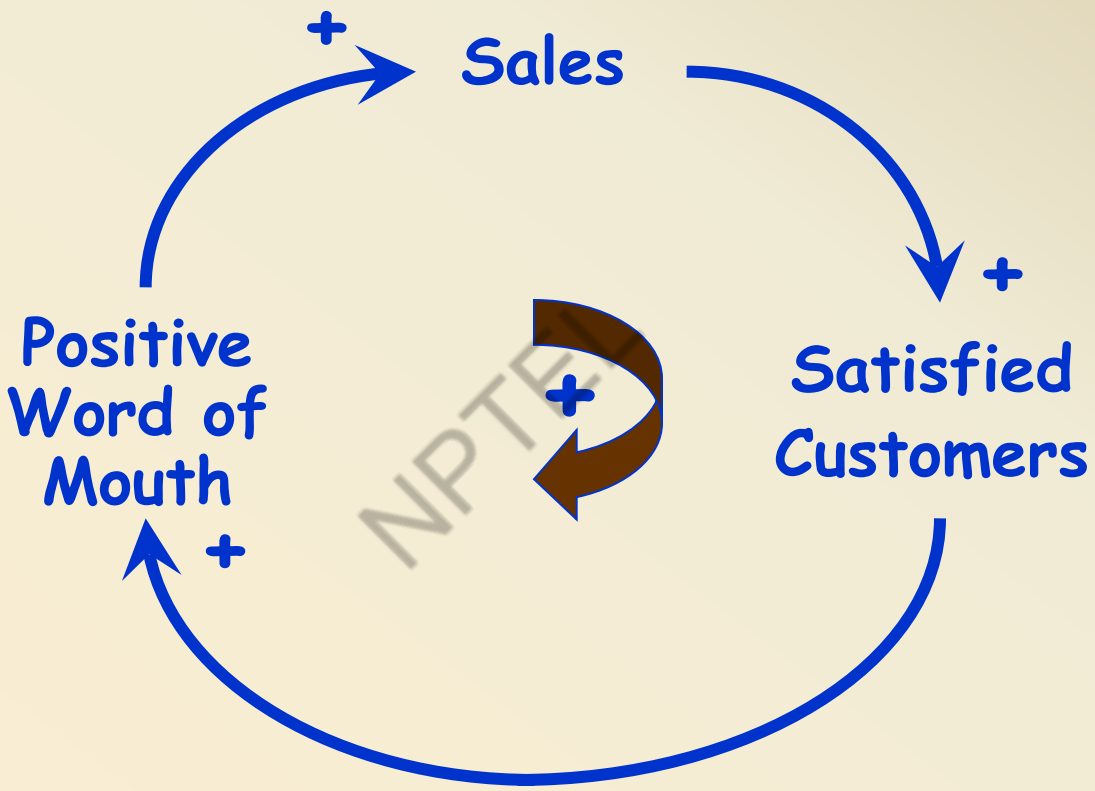
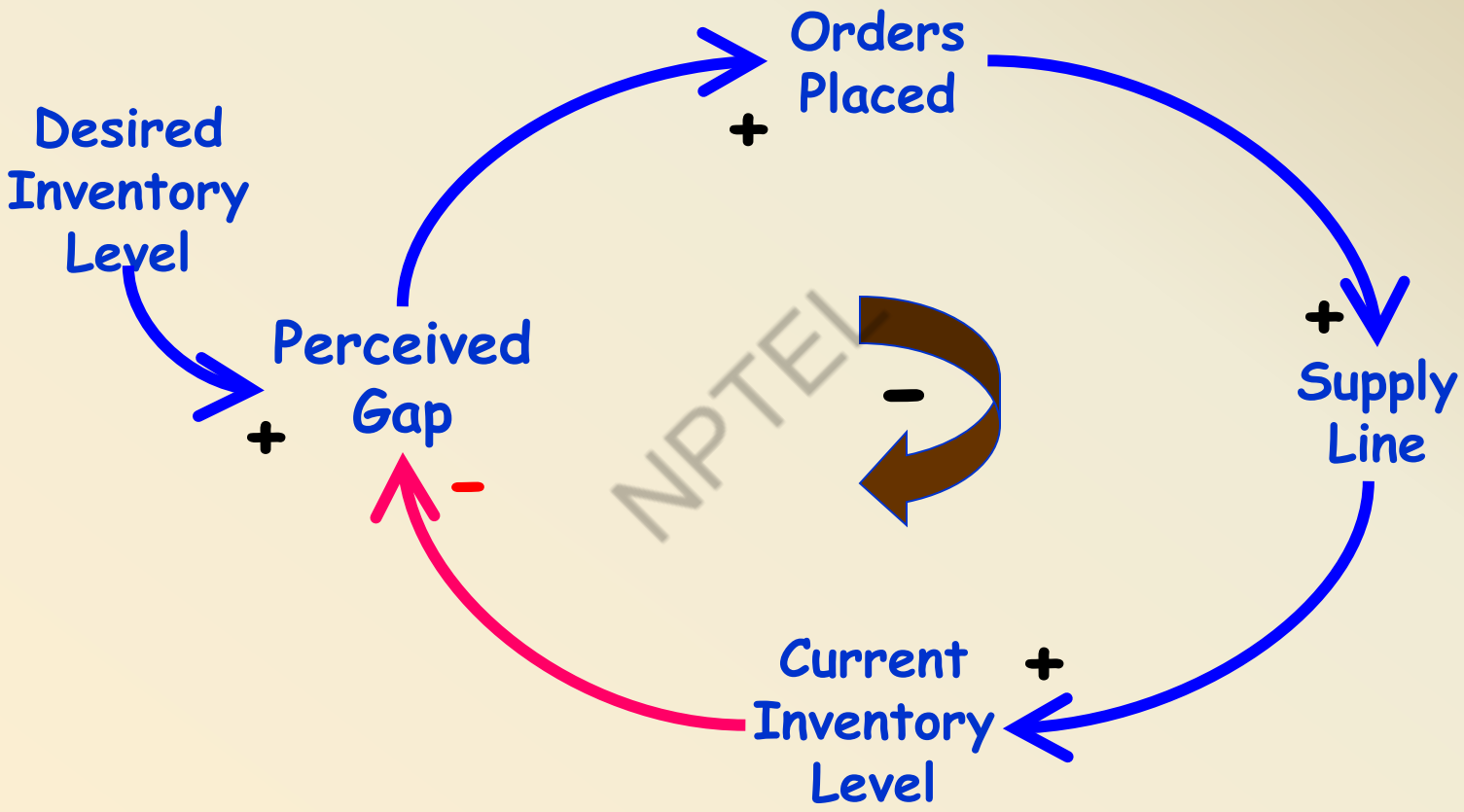
Positive Feedback



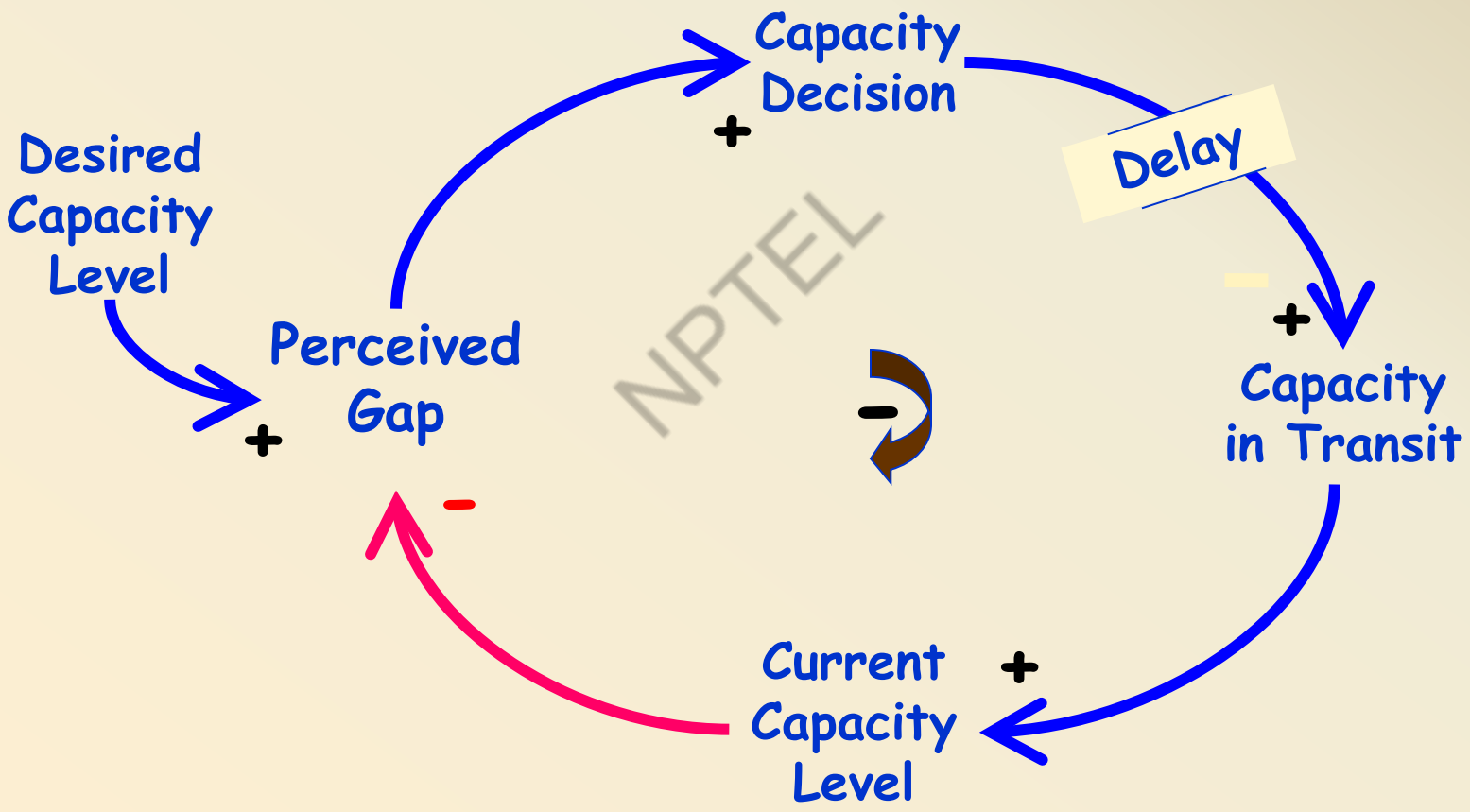
Sales continue to grow resulting from Positive Word of Mouth from Satisfied Customers

Negative Feedback



Balancing Process for Adjusting Inventory to Desired Inventory level using Perceived Gap as Controller

Feedback Loop With Delay



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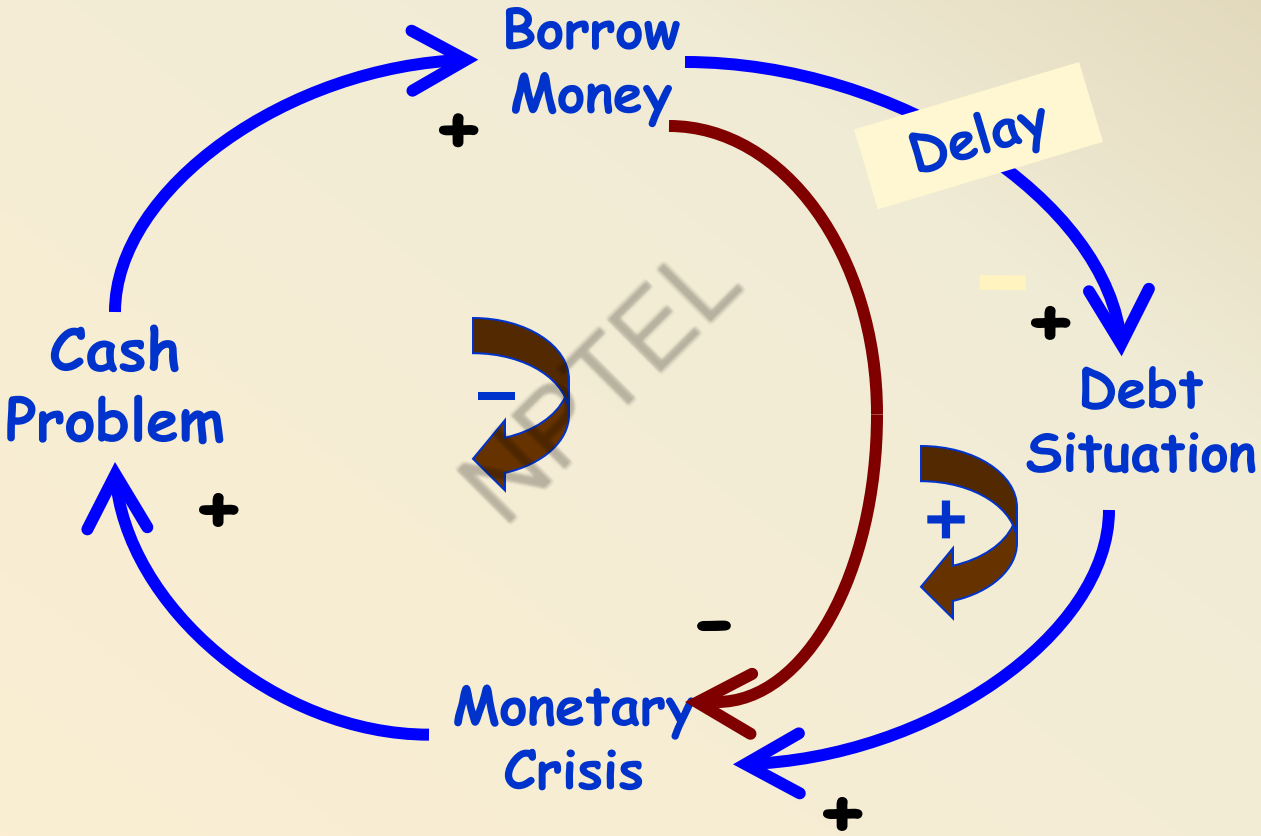
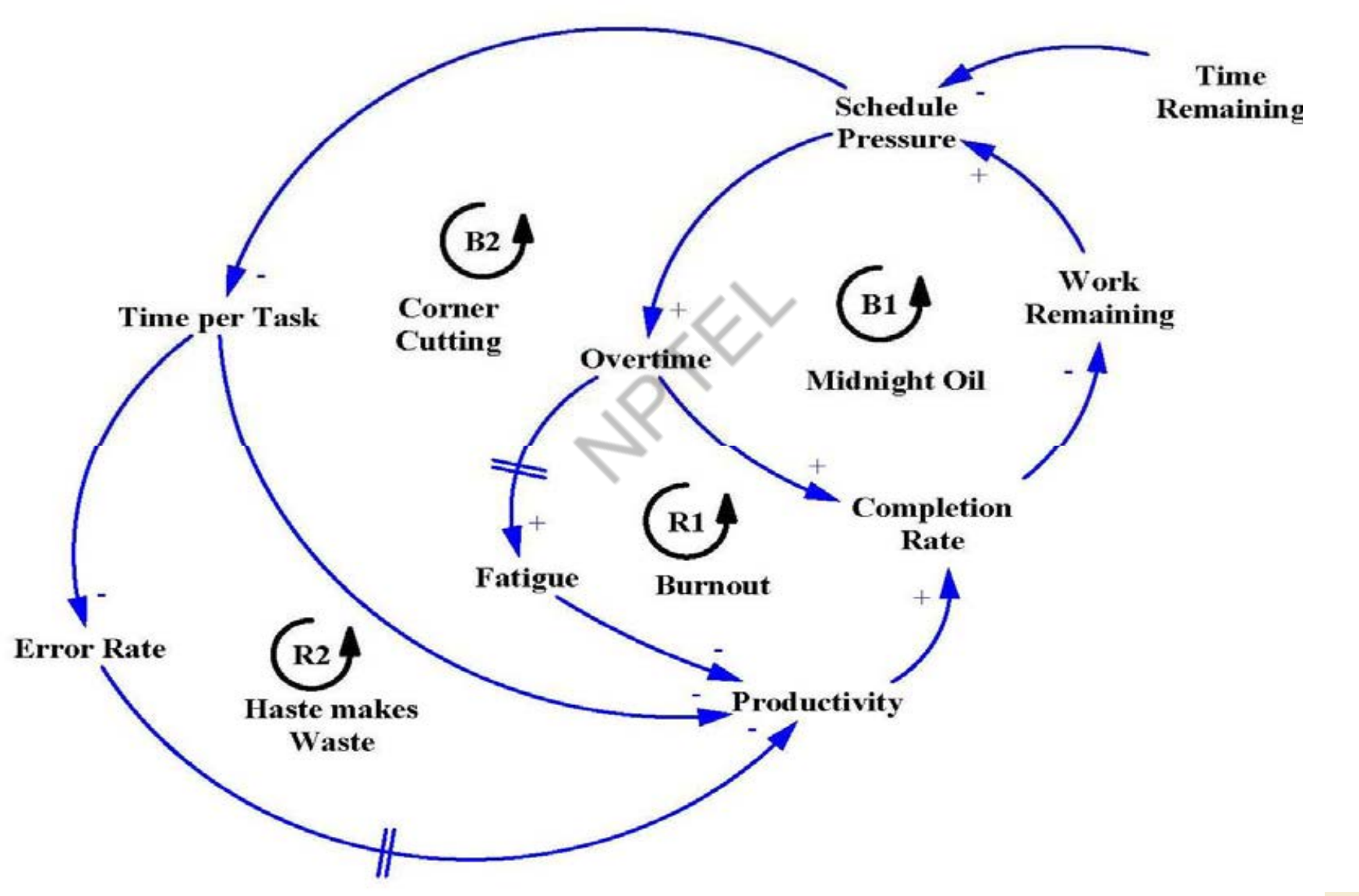
Fixes that Fail



Borrow Money to Make Payments

Schedule Pressure Dynamics



Axioms and Purpose

Two Axioms of System Dynamics

System Structure is made up of Feedback Loops

System Behavior is a function of System Structure and Policies

Hence the purpose of System Dynamics is to:

Analyze how the interaction between SYSTEM STRUCTURE and POLICIES determine the SYSTEM BEHAVIOR

Study system behaviors taking into account COMPLEX STRUCTURES of FEEDBACKS and TIME DELAYS

Use COMPUTER SIMULATION effectively to generate SYSTEM BEHAVIOR

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System Dynamics Example

The production process of a plant uses a certain raw material. Exogenous influences determine the usage rate of this material.

To ensure continuous production, the company keeps an inventory of the raw material. A supply delay of 5 months occurs between the ordering of additional raw material and its arrival in the inventory.

In an attempt to guard against inventory shortages, the management has decided to employ the formal policy of ordering exactly the amount of material consumed, i.e. order rate = usage rate.

Assume the exogenously determined usage rate increases linearly with time.

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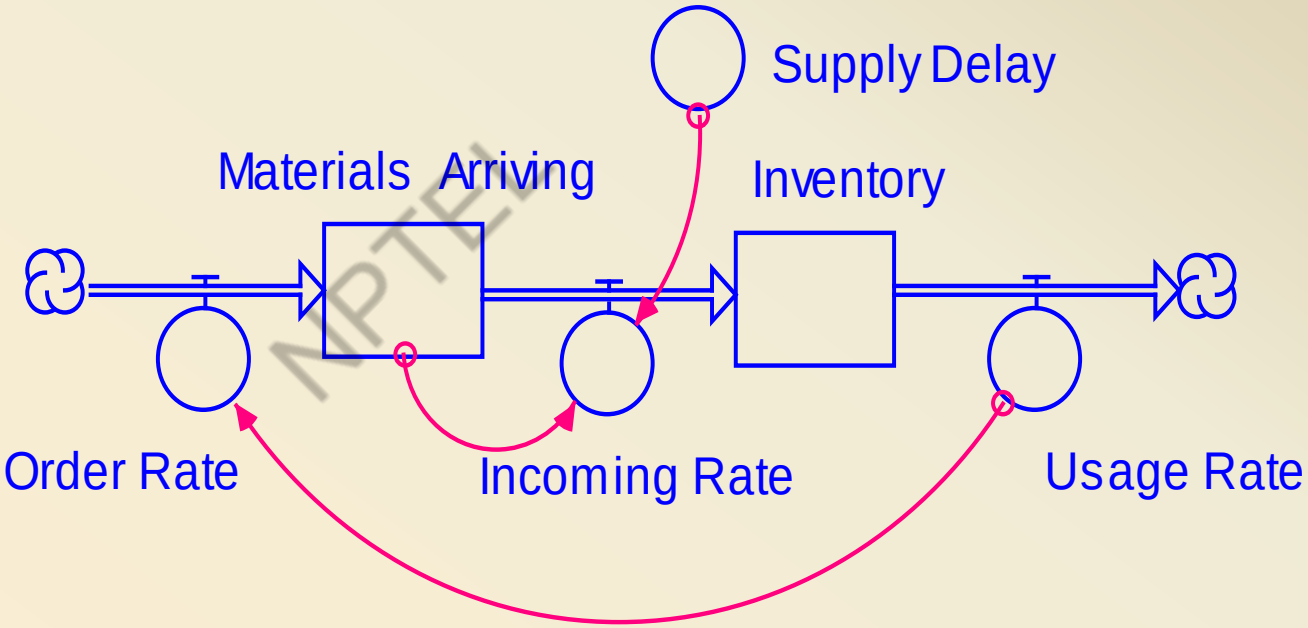
System Dynamics Example

Flow Diagram



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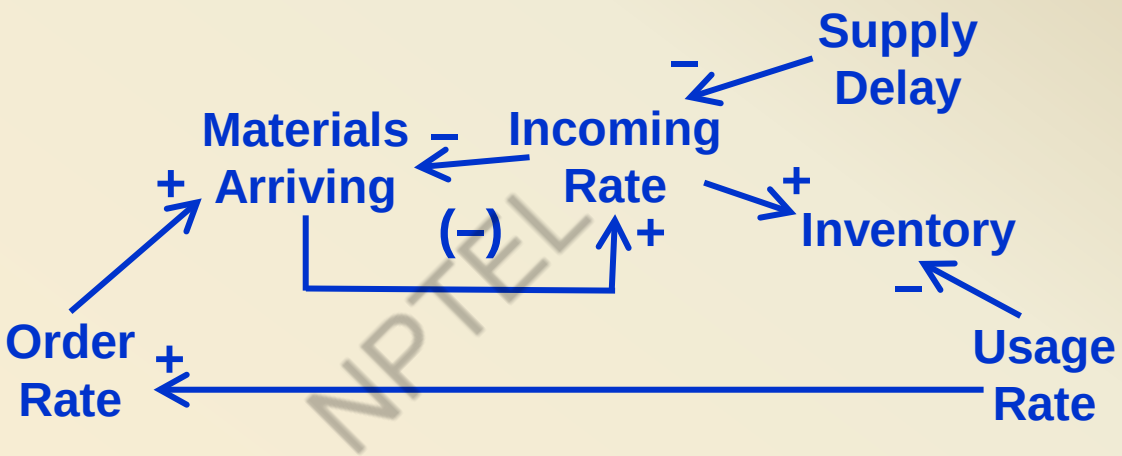
System Dynamics Example

Causal Loop Diagram



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System Dynamics Equations

- Inventory(t+dt) = Inventory(t) + (Incoming_Rate - Usage_Rate) * dt
- INIT Inventory = 100
- INFLOWS:
- Incoming_Rate = Materials__Arriving/Supply_Delay
- OUTFLOWS:
- Usage_Rate = 20+TIME*1
- Materials__Arriving(t+dt) = Materials__Arriving(t) + (Order_Rate - Incoming_Rate) * dt
- INIT Materials__Arriving = 100
- INFLOWS:
- Order_Rate = Usage_Rate
- OUTFLOWS:
- Incoming_Rate = Materials__Arriving/Supply_Delay
- Supply_Delay = 5

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System Dynamics Equations

Starting with a non-zero inventory, how will the *formal policy* work out when faced with growth in usage? Include a *sketch* of inventory over time in your answer.

Answer:

By using formal policy, inventory exponentially decays with time.

As usage rate grows continuously, order rate also increases similarly.

But incoming rate is delayed by 5 months and hence it is always lower than order rate and hence usage rate.

Hence, inventory continues to decay all the time.

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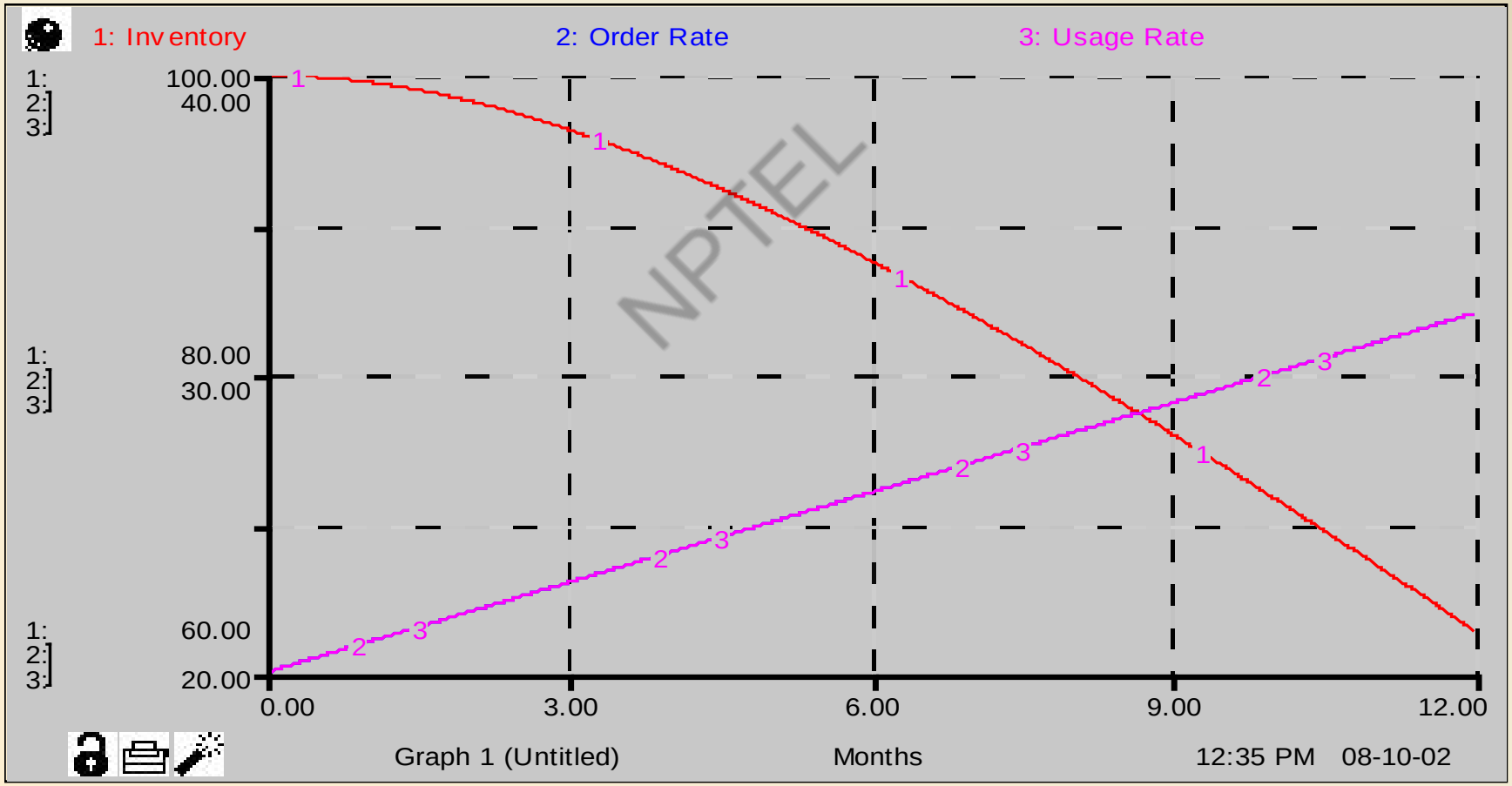
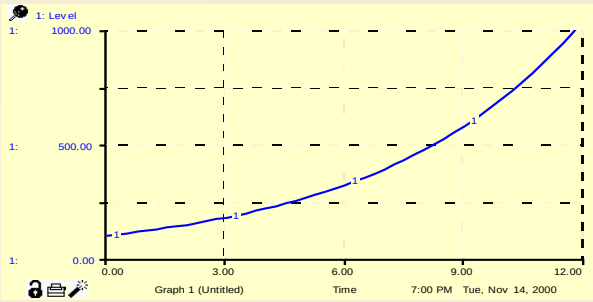
System Dynamics Equations

Starting with a non-zero inventory, how will the *formal policy* work out when faced with growth in usage? Include a *sketch* of inventory over time in your answer.

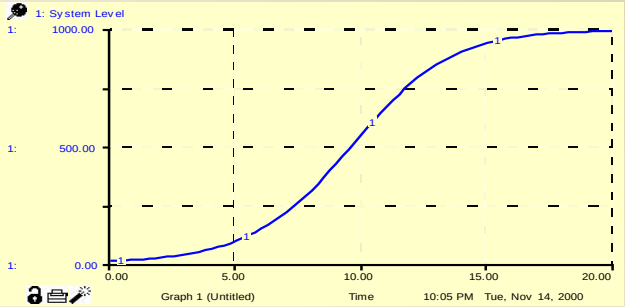


Basic System Behaviour

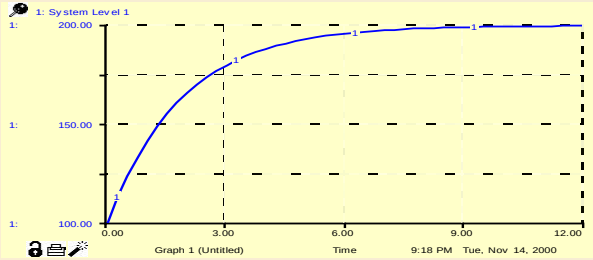
Exponential Growth



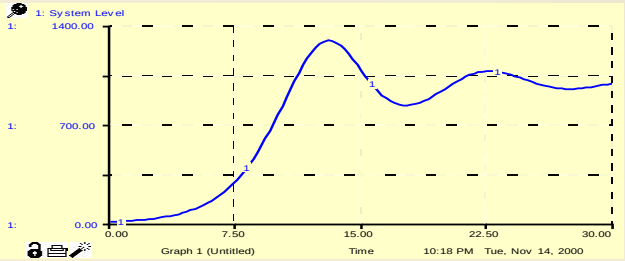
S-shaped Growth



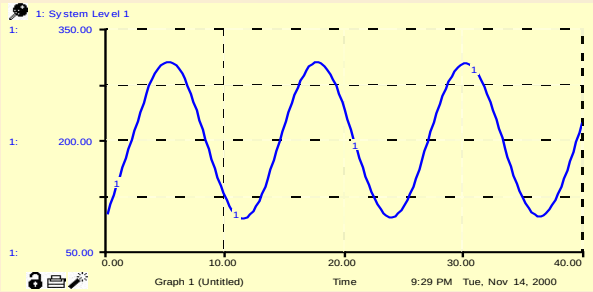
Goal Seeking



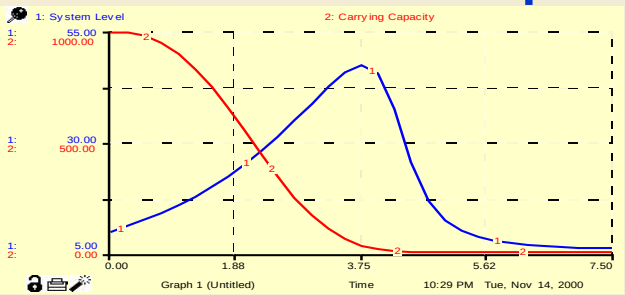
S-shaped Growth with Overshoot



Oscillations



Overshoot and Collapse



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THANK YOU

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